

PANEL 4. INTER PHASE SIMILARITY COMPUTATION

A. Centroid Normalization

Normalize to unit length

$$\hat{c}_1 = \frac{c_1}{\|c_1\|}$$

Apply to all centroids

$$\hat{c}_2 = \frac{c_2}{\|c_2\|}$$

$$\hat{c}_3 = \frac{c_3}{\|c_3\|}$$

Cosine similarity $\cos(\theta) = \hat{c}_i \cdot \hat{c}_j$

B. Inter Phase Similarity

Similarity between phase pairs

Similarity Matrix

	P1	P2	P3
P1	1.00	.65	.58
P2	.65	1.00	.72
P3	.58	.72	1.00

Mean similarity $\mu_{\text{inter}} = 0.65$